Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
DSIP(01CT1513)	**Aim :-** Design Butterworth and Chebyshev filter using bilinear transformation method.	
Experiment :- 4	**11-10-2026**	**Enrolment No:92400133131**

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import butter, bilinear, freqz, cheby1


def design_butterworth_filter(filter_order, cutoff_frequency,
sampling_frequency):
    # Design the analog Butterworth filter
    analog_b, analog_a = butter(filter_order, cutoff_frequency, analog=True,
btype='low')
    # Perform the bilinear transformation
    digital_b, digital_a = bilinear(analog_b, analog_a, sampling_frequency)
    return digital_b, digital_a


def design_chebyshev_filter(filter_order, cutoff_frequency,
sampling_frequency, ripple):
    # Design the analog Chebyshev filter
    analog_b, analog_a = cheby1(filter_order, ripple, cutoff_frequency,
analog=True, btype='low')
    # Perform the bilinear transformation
    digital_b, digital_a = bilinear(analog_b, analog_a, sampling_frequency)
    return digital_b, digital_a


def plot_filter_response(digital_b, digital_a, sampling_frequency):
```

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```
    # Compute the frequency response of the filter
    frequency, magnitude_response = freqz(digital_b, digital_a,
fs=sampling_frequency)

    # Plot the magnitude response
    plt.figure(figsize=(10, 6))
    plt.plot(frequency, np.abs(magnitude_response))
    plt.title('Filter Magnitude Response')
    plt.xlabel('Frequency (Hz)')
    plt.ylabel('Magnitude')
    plt.grid(True)
    plt.show()

    # Compute the impulse response of the filter
    _, impulse_response = freqz(digital_b, digital_a, fs=sampling_frequency,
worN=4096)

    # Plot the impulse response
    plt.figure(figsize=(10, 6))
    plt.plot(impulse_response)
    plt.title('Filter Impulse Response')
    plt.xlabel('Samples')
    plt.ylabel('Amplitude')
    plt.grid(True)
    plt.show()


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# Department of Information and Communication Technology

# Specify the desired filter specifications
filter_order = 4  # Filter order
cutoff_frequency = 1000  # Cutoff frequency in Hz
sampling_frequency = 8000  # Sampling frequency in Hz
ripple = 0.5  # Ripple factor for Chebyshev filter

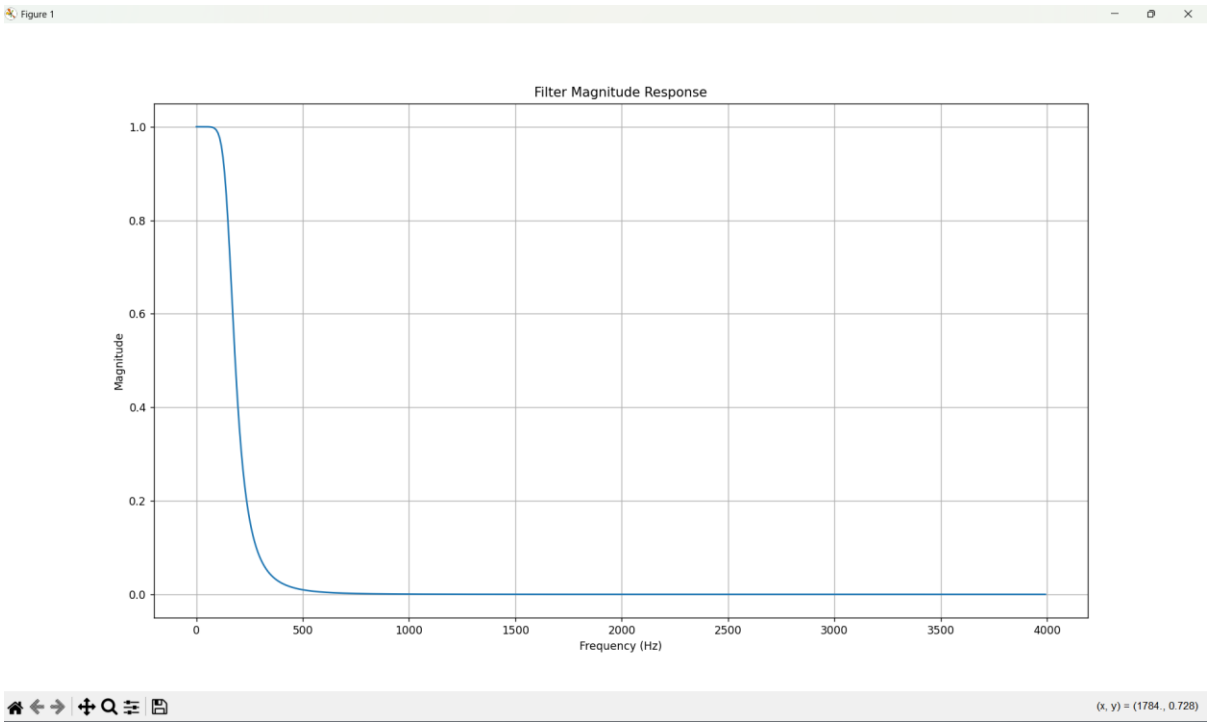
# Design the Butterworth filter
```

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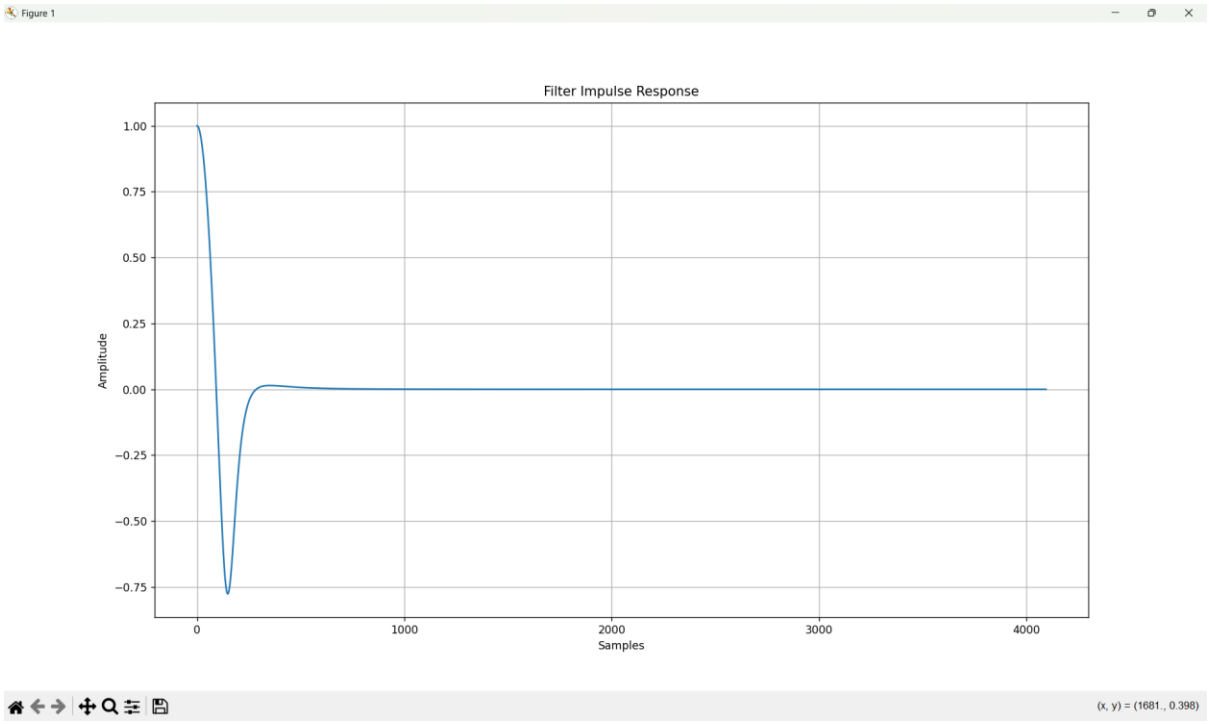
```
digital_b, digital_a = design_butterworth_filter(filter_order,
cutoff_frequency, sampling_frequency)
# Plot the Butterworth filter's magnitude response and impulse response
plot_filter_response(digital_b, digital_a, sampling_frequency)

# Design the Chebyshev filter
digital_b, digital_a = design_chebyshev_filter(filter_order, cutoff_frequency,
sampling_frequency, ripple)
# Plot the Chebyshev filter's magnitude response and impulse response
plot_filter_response(digital_b, digital_a, sampling_frequency)

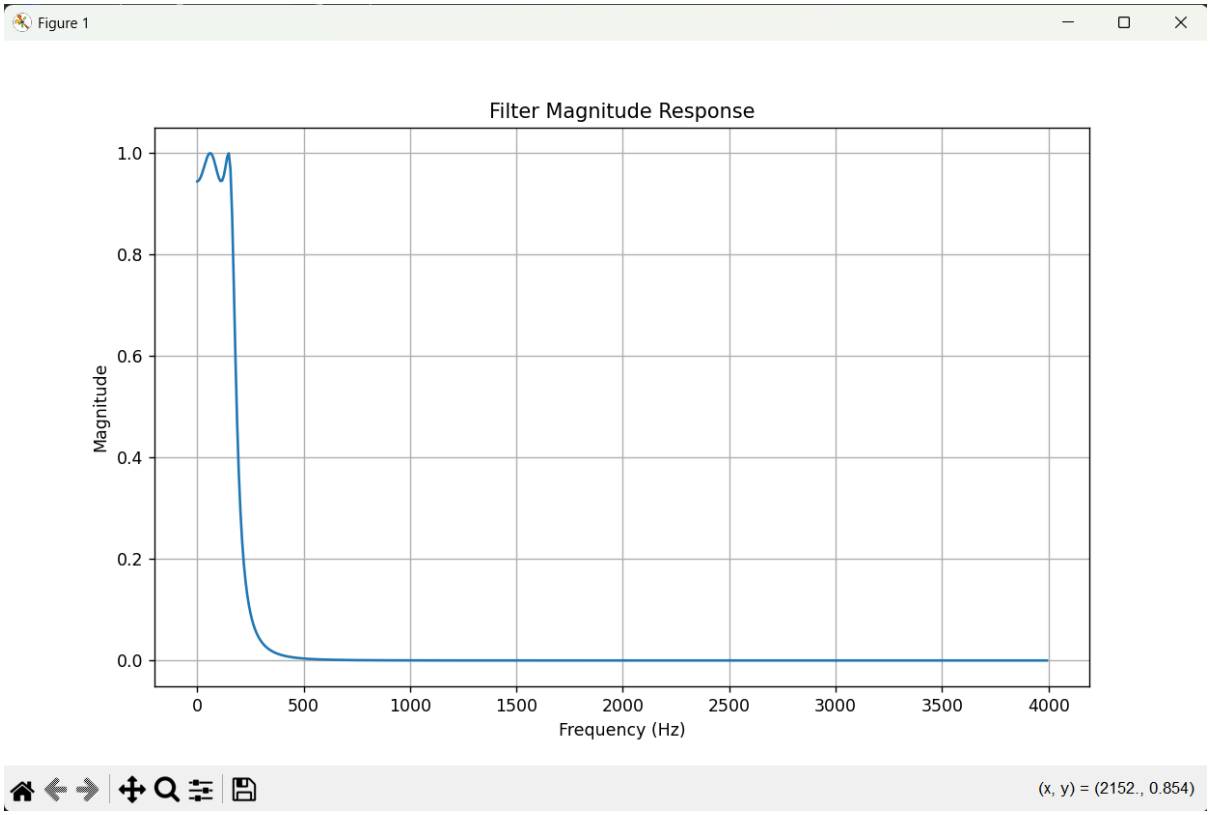
# Save the filter coefficients (optional)
filter_path = 'filter_coefficients.txt'
np.savetxt(filter_path, np.vstack((digital_b, digital_a)), delimiter=',')
print(f"Filter coefficients saved at: {filter_path}")
```



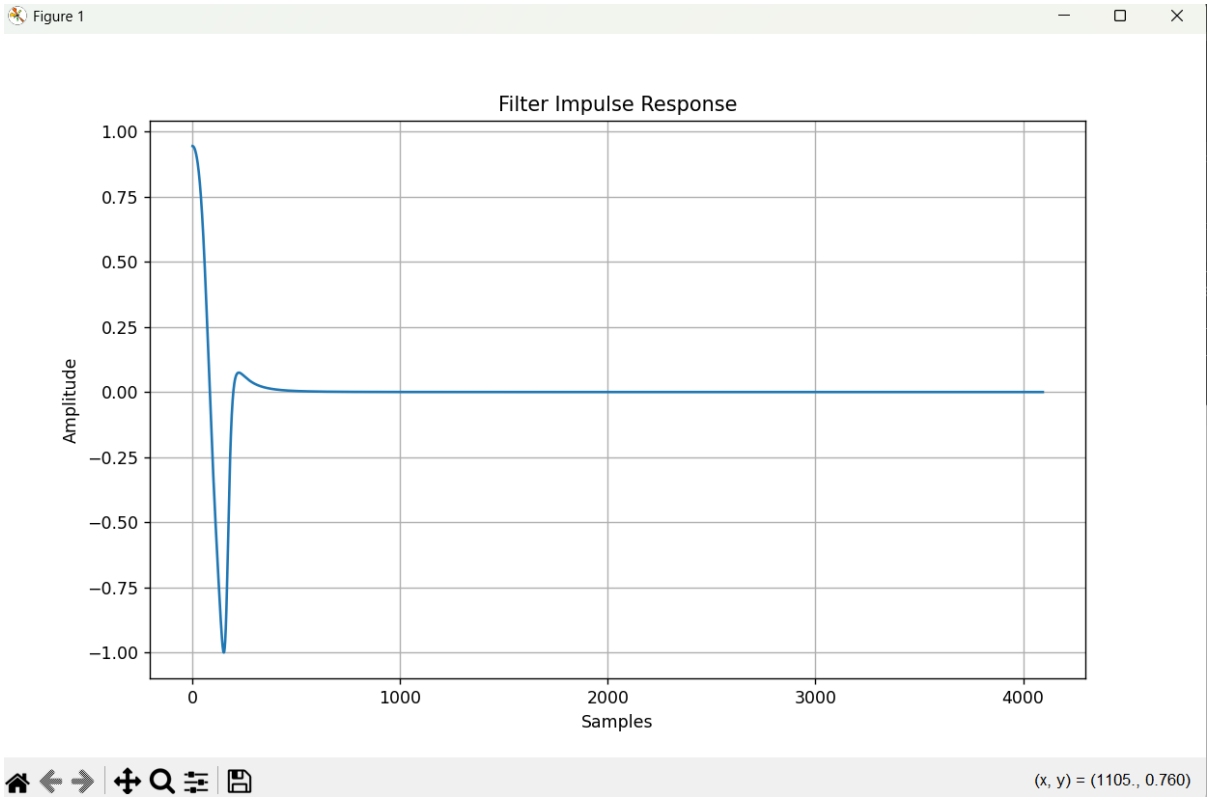
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
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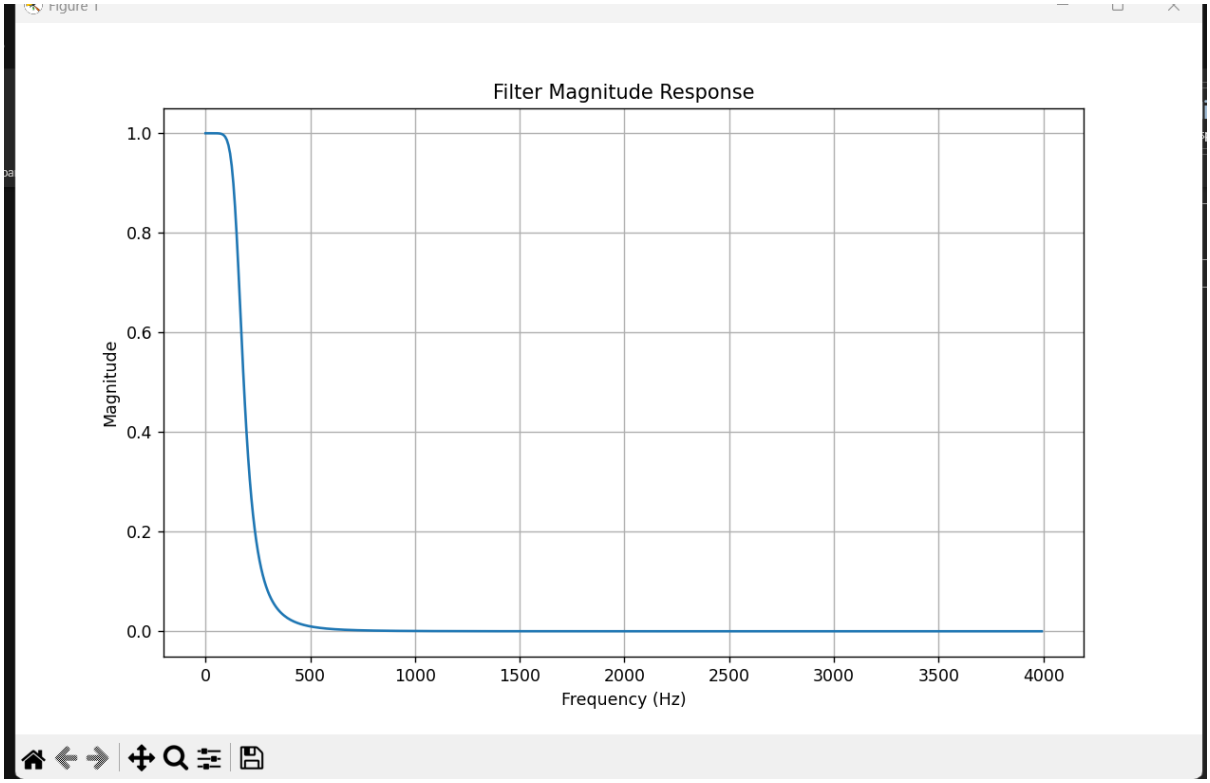


Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
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Additional Task

•Generate a discrete sine wave of 1KHz, 2KHz, 3KHz, 4 KHz. Plot all these

CODE :-
```
import numpy as np
import matplotlib.pyplot as plt

fs = 10000
t = np.arange(0, 0.01, 1/fs)
x1 = np.sin(2*np.pi*1000*t)
x2 = np.sin(2*np.pi*2000*t)
x3 = np.sin(2*np.pi*3000*t)
x4 = np.sin(2*np.pi*4000*t)

plt.figure(figsize=(10, 8))
plt.subplot(4, 1, 1)
plt.stem(t, x1)
plt.title("1 kHz Sine Wave")
plt.grid()

plt.subplot(4, 1, 2)
plt.stem(t, x2)
plt.title("2 kHz Sine Wave")
plt.grid()

plt.subplot(4, 1, 3)
plt.stem(t, x3)
plt.title("3 kHz Sine Wave")
plt.grid()

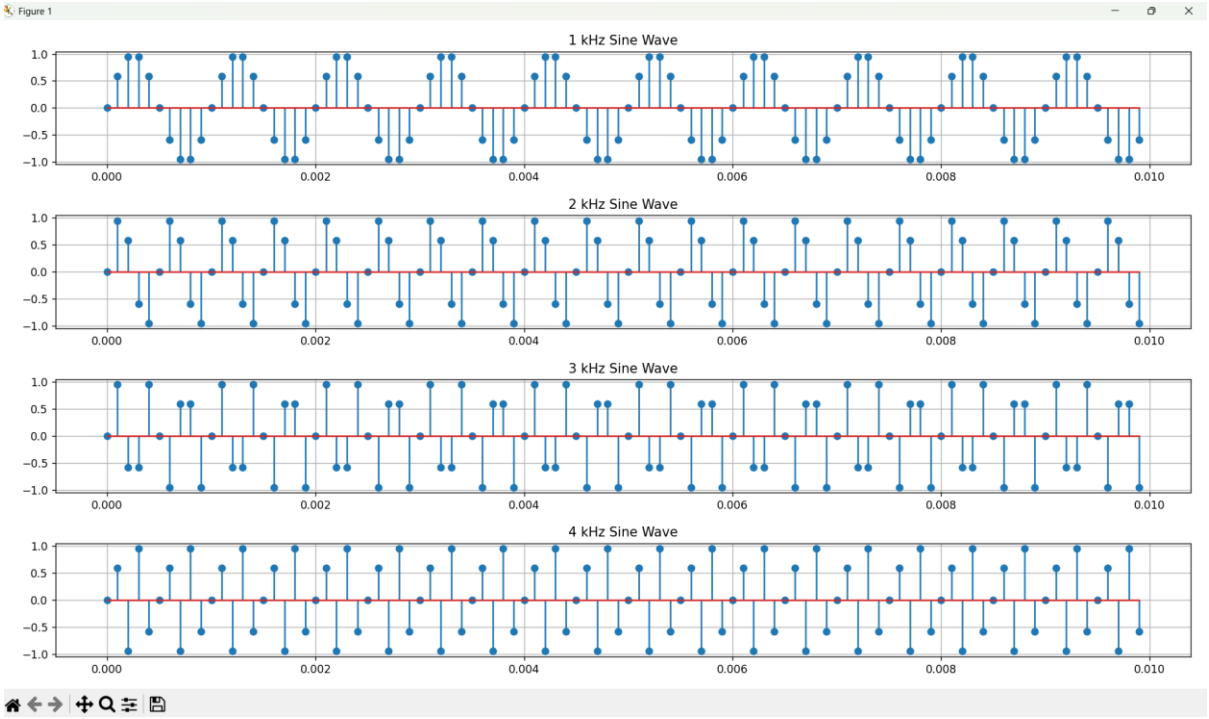
plt.subplot(4, 1, 4)
plt.stem(t, x4)
plt.title("4 kHz Sine Wave")
plt.grid()

plt.tight_layout()
```

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```
plt.show()
```

OUTPUT :-



•Add all these four signal and plot it in time as well as frequency domain

CODE :-

```
import numpy as np
import matplotlib.pyplot as plt

fs = 10000
t = np.arange(0, 0.01, 1 / fs)

x1 = np.sin(2 * np.pi * 1000 * t)
x2 = np.sin(2 * np.pi * 2000 * t)
x3 = np.sin(2 * np.pi * 3000 * t)
x4 = np.sin(2 * np.pi * 4000 * t)
x = x1 + x2 + x3 + x4

# Time domain
plt.figure(figsize=(10, 4))
plt.stem(t, x)
```

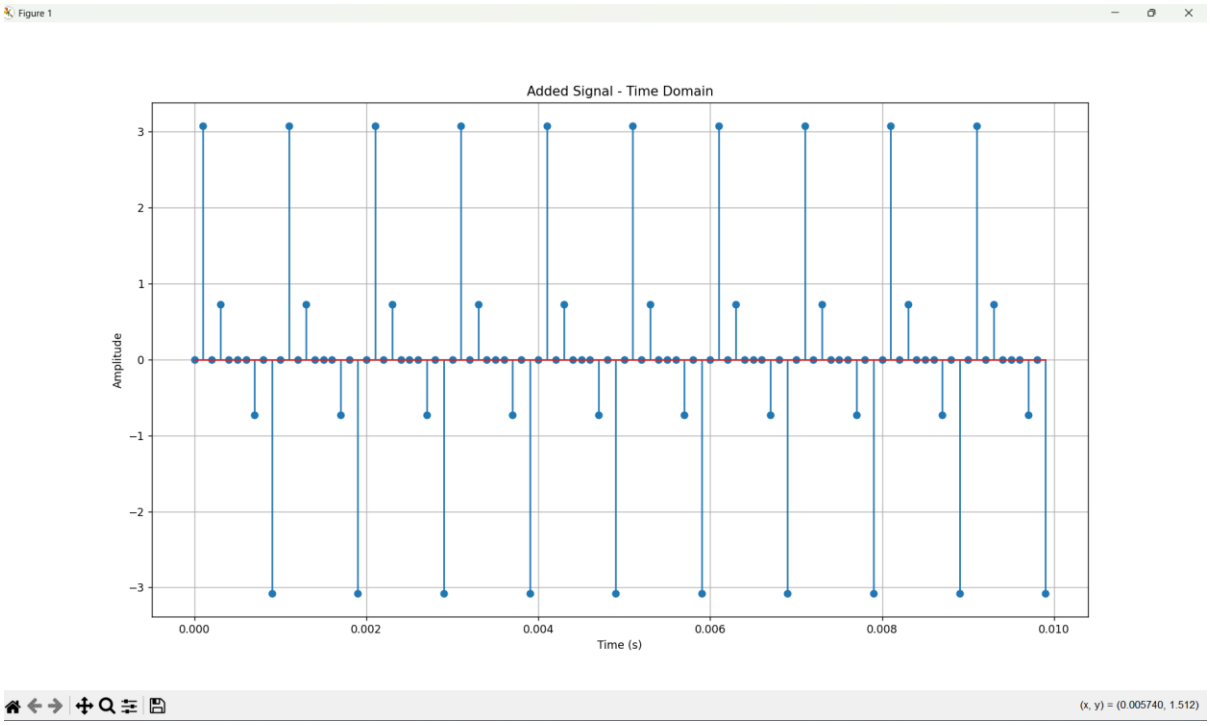
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
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```
plt.title("Added Signal - Time Domain")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid()
plt.show()

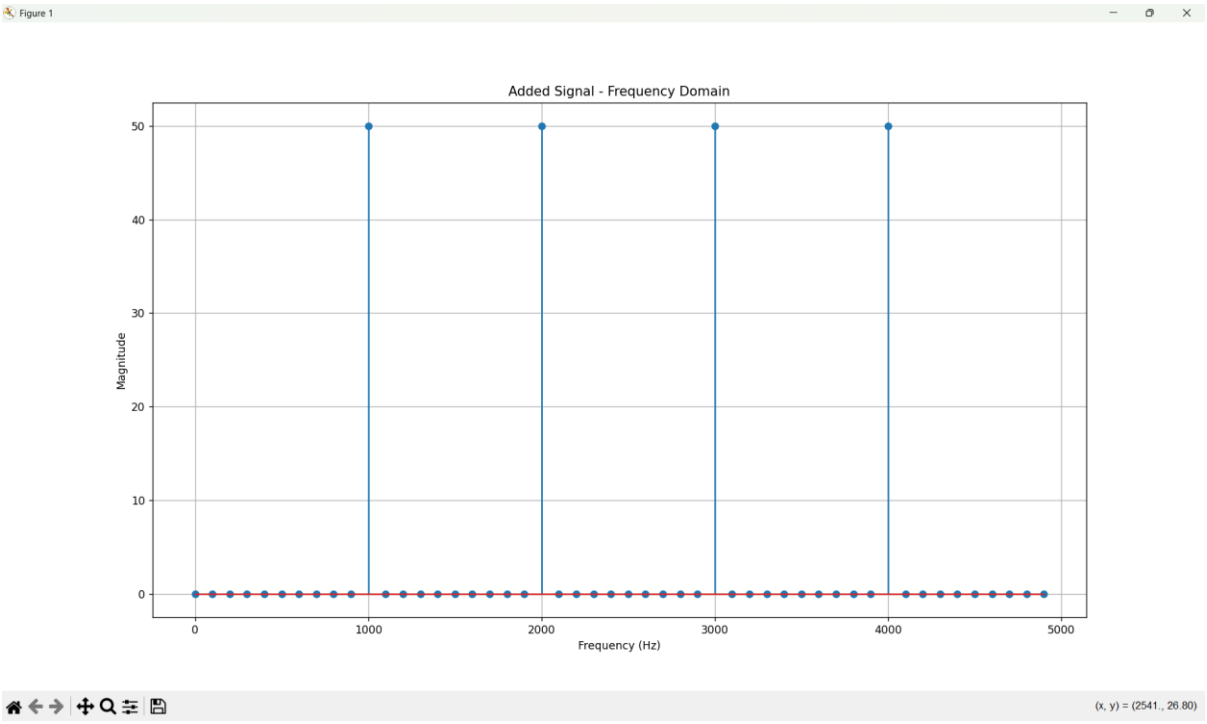
# Frequency domain
X = np.fft.fft(x)
freq = np.fft.fftfreq(len(x), 1 / fs)

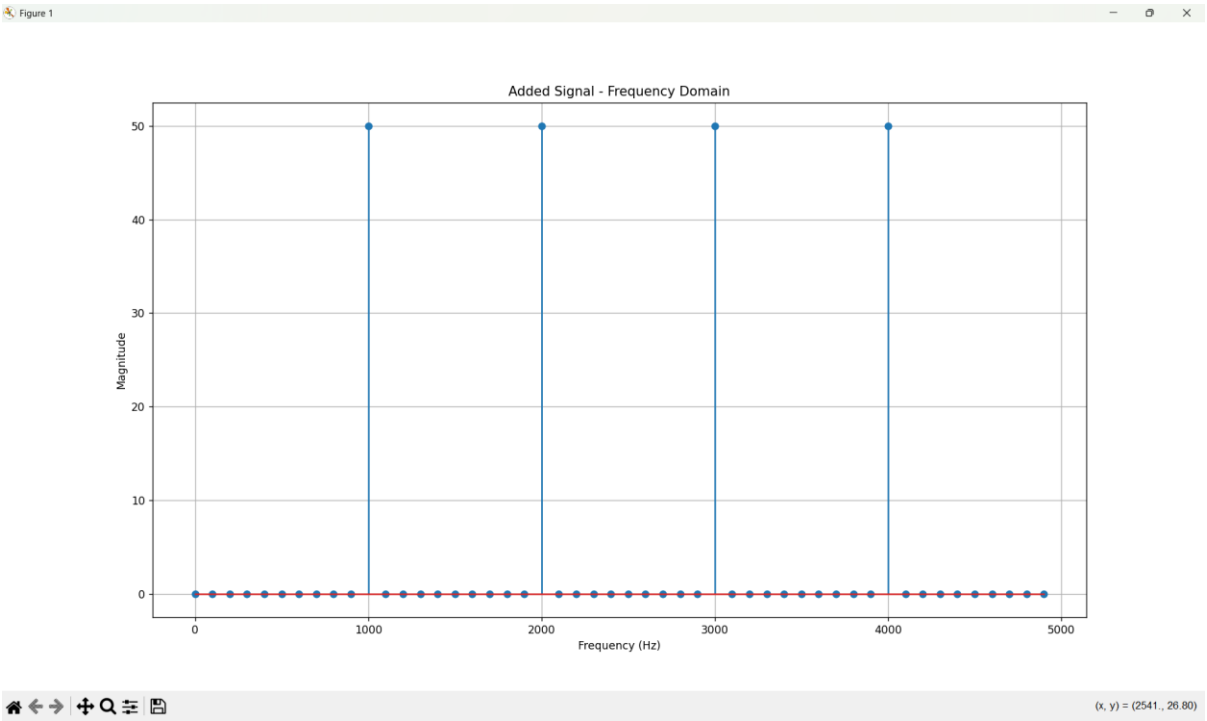
plt.figure(figsize=(10, 4))
plt.stem(freq[:len(freq) // 2], np.abs(X[:len(X) // 2]))
plt.title("Added Signal - Frequency Domain")
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.grid()
plt.show()
```

OUTPUT :-



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• pass through buttorworth digital filter having cutoff frequency 2 khz and plot the output plot in time as well as frequency domain

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CODE :-

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import butter, filtfilt

fs = 10000
t = np.arange(0, 0.01, 1/fs)

x1 = np.sin(2*np.pi*1000*t)
x2 = np.sin(2*np.pi*2000*t)
x3 = np.sin(2*np.pi*3000*t)
x4 = np.sin(2*np.pi*4000*t)
x = x1 + x2 + x3 + x4

# Butterworth filter
b, a = butter(4, 2000/(fs/2), btype='low')
y = filtfilt(b, a, x)

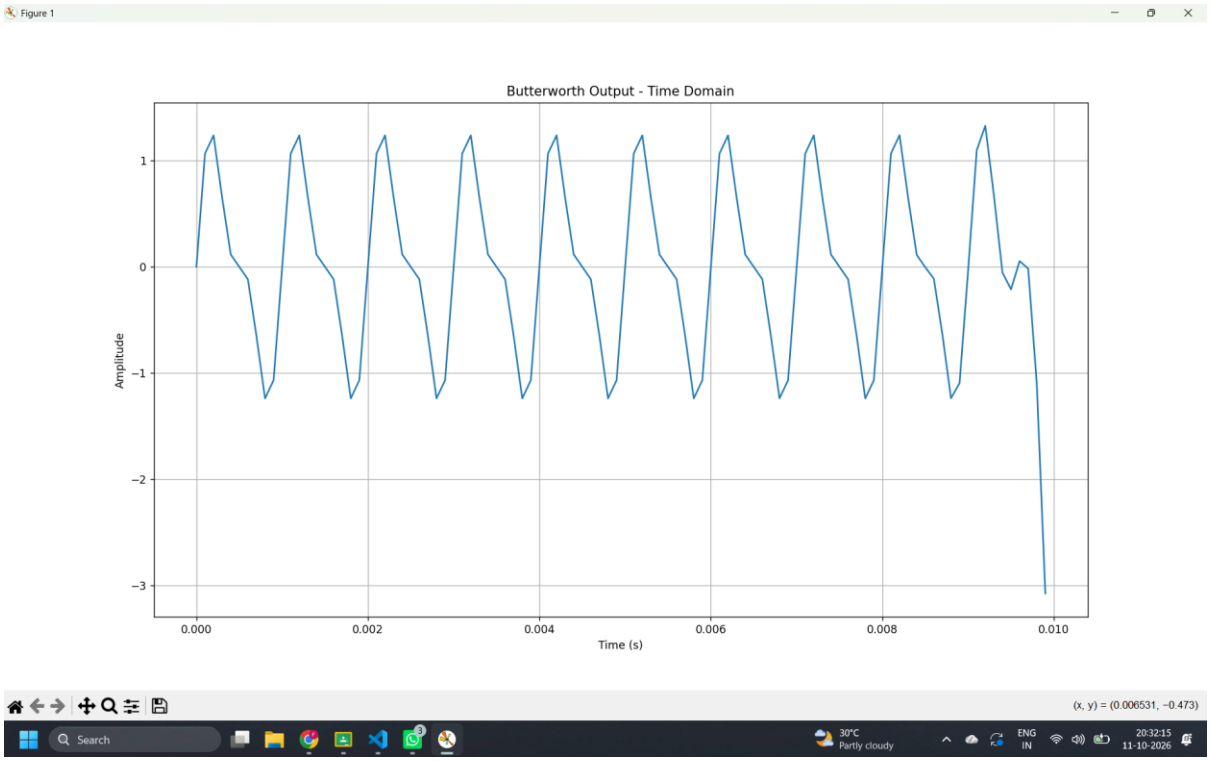
# Time domain
plt.figure(figsize=(10, 4))
plt.plot(t, y)
plt.title("Butterworth Output - Time Domain")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid()
plt.show()

# Frequency domain
Y = np.fft.fft(y)
freq = np.fft.fftfreq(len(y), 1/fs)

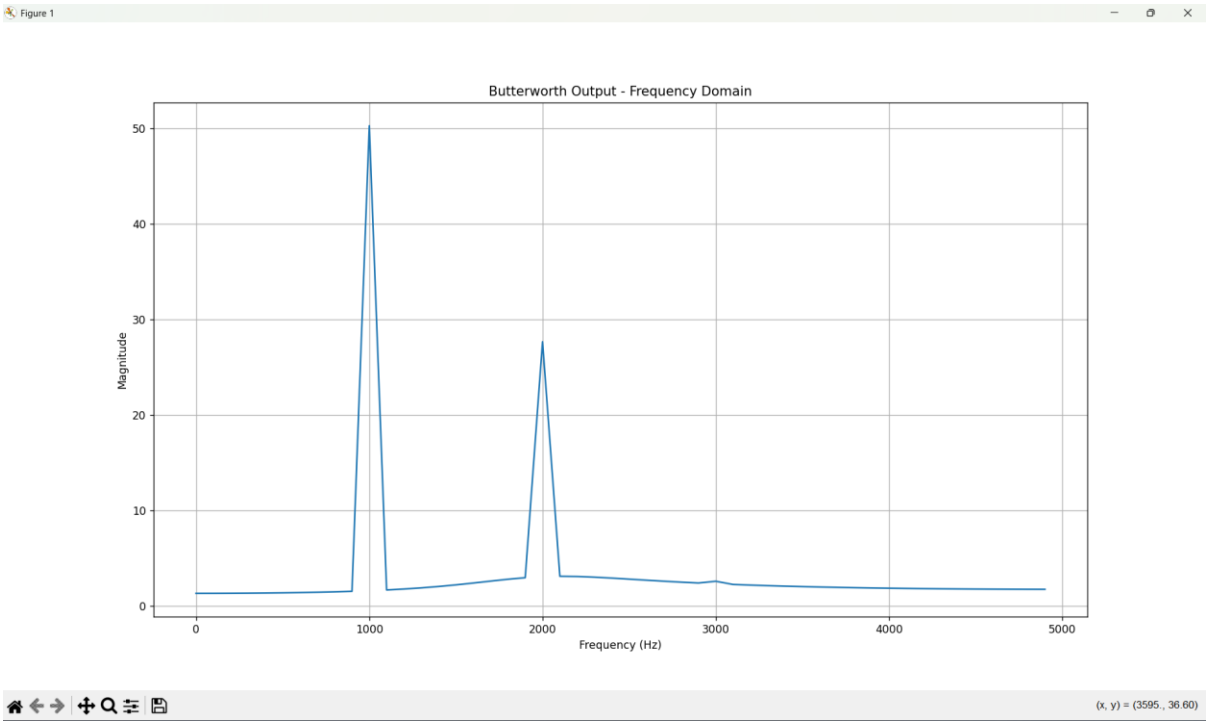
plt.figure(figsize=(10, 4))
plt.plot(freq[:len(freq)//2], np.abs(Y[:len(Y)//2]))
plt.title("Butterworth Output - Frequency Domain")
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.grid()
plt.show()
```

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OUTPUT ;-



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•Pass through chebyschev digital filter having cutoff frequency 3KHz and plot the output plot in time as well as frequency domain

CODE :-
```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import cheby1, filtfilt

fs = 10000
t = np.arange(0, 0.01, 1/fs)
x1 = np.sin(2*np.pi*1000*t)
x2 = np.sin(2*np.pi*2000*t)
x3 = np.sin(2*np.pi*3000*t)
x4 = np.sin(2*np.pi*4000*t)
x = x1 + x2 + x3 + x4

# Chebyshev filter
b, a = cheby1(4, 0.5, 3000/(fs/2), btype='low')
y = filtfilt(b, a, x)

# Time domain
plt.figure(figsize=(10, 4))
```

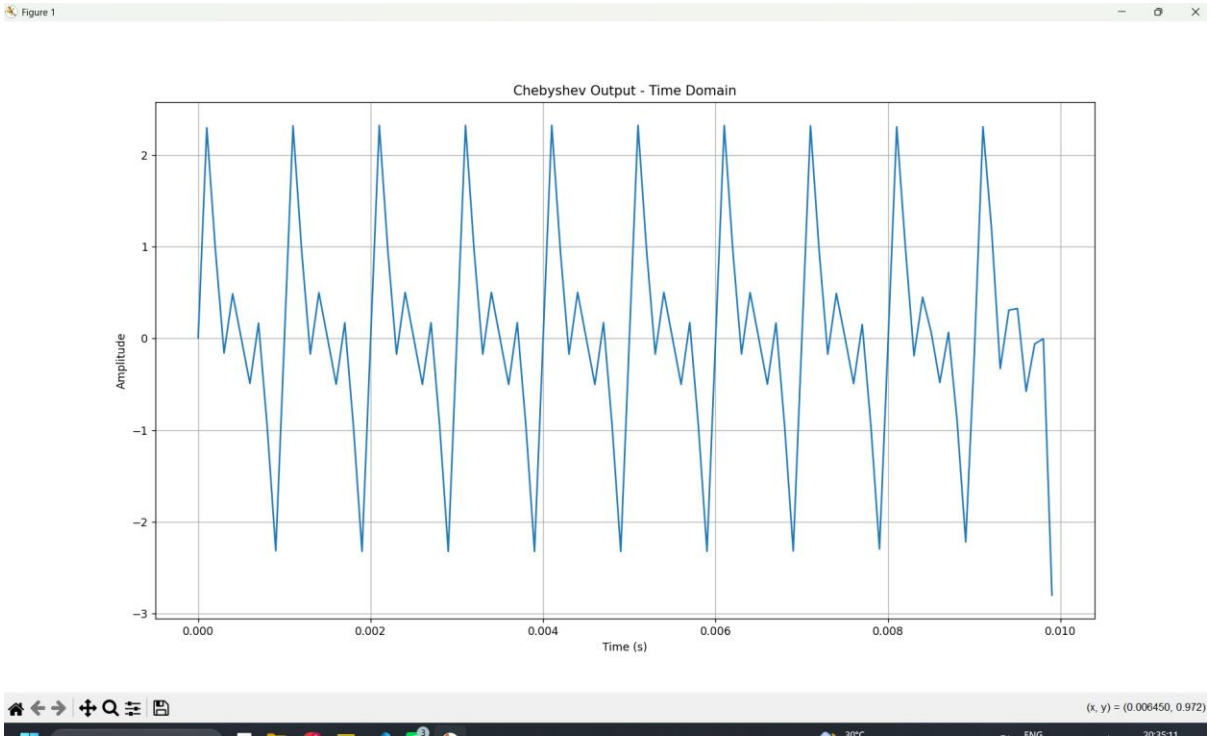
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```
plt.plot(t, y)
plt.title("Chebyshev Output - Time Domain")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid()
plt.show()

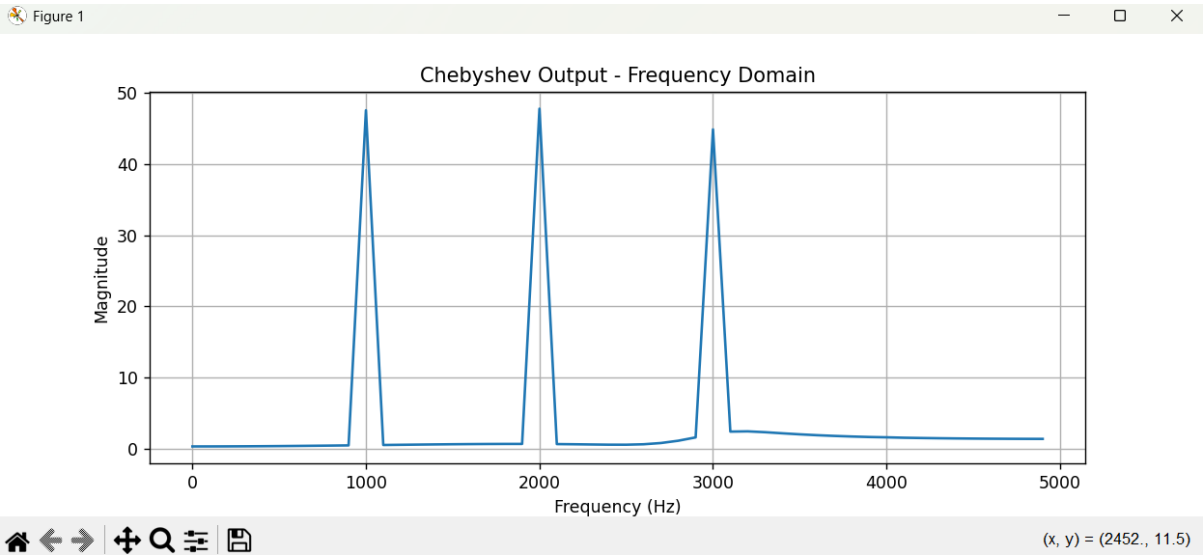
# Frequency domain
Y = np.fft.fft(y)
freq = np.fft.fftfreq(len(y), 1/fs)

plt.figure(figsize=(10, 4))
plt.plot(freq[:len(freq)//2], np.abs(Y[:len(Y)//2]))
plt.title("Chebyshev Output - Frequency Domain")
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.grid()
plt.show()
```

OUTPUT :-



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- CONCLUSION :-The Butterworth and Chebyshev low-pass filters were successfully designed using Python.
- The NumPy library was used for numerical calculations.
- The SciPy library was used to design and analyze the digital filters.
- The Butterworth filter was designed with an order of 4.
- The Chebyshev filter was designed with an order of 4 and a ripple factor of 0.5 dB.
- The cutoff frequency was set to 1000 Hz.
- The sampling frequency was set to 8000 Hz.
- The bilinear transformation was used to convert analog filters into digital filters.
- The frequency response of both filters was calculated using the `freqz()` function.
- The magnitude response graphs were plotted using Matplotlib.
- The filters allow low-frequency signals to pass while attenuating high-frequency signals.
- The Butterworth filter provides a smooth frequency response in the passband.
- The Chebyshev filter provides a sharper transition but introduces ripple in the passband.
- The filter coefficients were saved in a text file for future use.
- Thus, the experiment demonstrates the design, transformation, visualization, and analysis of digital low-pass filters using Python.

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